5 Joint Probability Distributions

5.1 Joint Probability Distributions for Two Random Variables

Definition. Let $X,\ Y$ be two random variables. With a ordered pair $(X,\ Y)$, we define a **joint probability distribution function** and denote

$$f_{XY}(x,\ y).$$

In discrete case, we called **joint probability mass function**; In continuous case, we called **joint probability density function**.

Property. Let $X,\ Y$ be two random variables.

1. If $f_{XY}(x,\ y)$ is a joint probability mass function, then it satisfies the following properties:

    (1) $f_{XY}(x,\ y) \geq 0$

    (2) $\displaystyle\sum_{x\in X}\sum_{y\in Y} f_{XY}(x,\ y) = 1$

    (3) $f_{XY}(x,\ y) = P(X = x,\ Y = y) = P(X = x\ \ \&\ \ Y = y)$

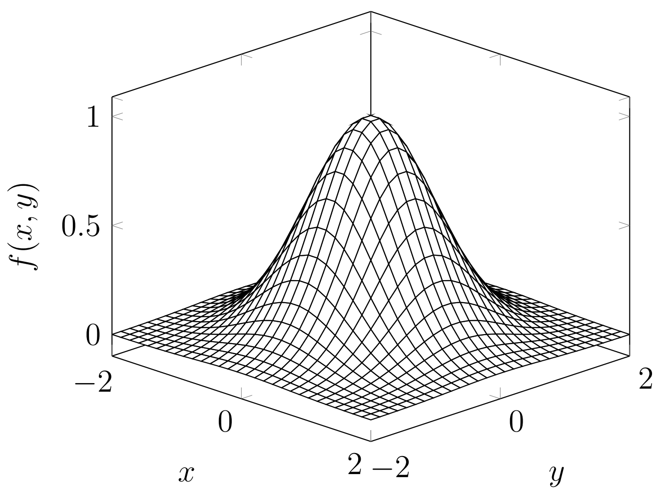
2. If $f_{XY}(x,\ y)$ is a joint probability density function, then it satisfies the following properties:

    (1) $f_{XY}(x,\ y) \geq 0$ for all $x,\ y$

    (2) $\displaystyle\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(x,\ y)\,dx\,dy = 1$

    (3) For any region R of two-dimensional space,

$$f_{XY}(x,\ y) = P((X,\ Y) \in \mathbb{R}) = \iint_R f_{XY}(x,\ y)\,dx\,dy.$$



Definition. (Marginal Density) Let $X,\ Y$ be two random variables.

1. For discrete case, the marginal(邊際) density is defined as

   (1) $f_X(x) = \sum_{y \in Y} f_{XY}(x,\ y)$

   (2) $f_Y(y) = \sum_{x \in X} f_{XY}(x,\ y)$

2. For continuous case, the marginal density is defined as

   (1) $f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,\ y)\, dy$

   (2) $f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,\ y)\, dx$

Example. Suppose $f_{XY}(x,\ y)$ is a joint probability mass function with two random variables $X = \{0,\ 1,\ 2\}$ and $Y = \{0,\ 1,\ 2,\ 3\}$. It is as shown in the table below

$Y = y$	$X = x$		
	0	1	2
0	0.15	0.1	0.05
1	0.02	0.1	0.05
2	0.02	0.03	0.2
3	0.01	0.02	0.25

Then the marginal probability distributions of X and Y are

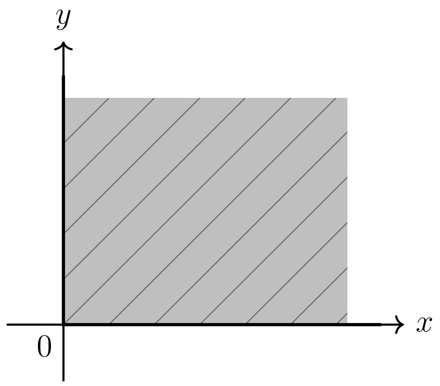
$Y = y$	$X = x$			
	0	1	2	$f_Y(y)$
0	0.15	0.1	0.05	**0.3**
1	0.02	0.1	0.05	**0.17**
2	0.02	0.03	0.2	**0.25**
3	0.01	0.02	0.25	**0.28**
$f_X(x)$	**0.2**	**0.25**	**0.55**	1

Example. Let $f_{XY}(x,\, y)$ be a joint probability density function and be defined as

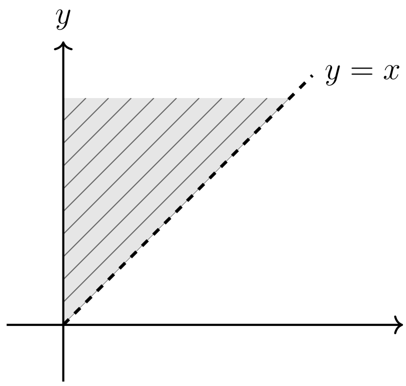
$$f_{XY}(x,\ y) = ke^{-0.01x-0.02y} \text{ for } 0 < x < y < \infty,$$

where $k = 6 \times 10^{-6}$. The region with nonzero probability is shaded in below. Find the $P(Y > 2000)$.

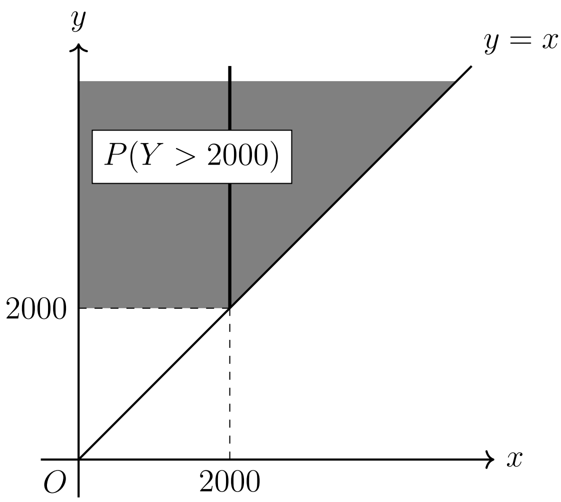
(i) **Region :** $0 < x < \infty,\ 0 < y < \infty$



(ii) **Region :** $0 < x < y < \infty$



Sol. The region is shaded in below.



So

$$
\begin{aligned}
P(Y > 2000) &= \int_{2000}^{\infty} \int_{0}^{y} f_{XY}(x, y)\, dx\, dy \\
&= \int_{2000}^{\infty} \int_{0}^{y} k e^{-0.01x - 0.02y}\, dx\, dy \\
&= k \int_{2000}^{\infty} e^{-0.02y} \left[\int_{0}^{y} e^{-0.01x}\, dx \right] dy \\
&= k \int_{2000}^{\infty} e^{-0.02y} \left[\frac{e^{-0.01x}}{-0.01} \right]_{0}^{y} dy \\
&= k \int_{2000}^{\infty} e^{-0.02y} \cdot 100(1 - e^{-0.01y})\, dy \\
&= 100k \int_{2000}^{\infty} (e^{-0.02y} - e^{-0.03y})\, dy \\
&= 100k \lim_{b\to\infty} \left[\frac{e^{-0.02y}}{-0.02} - \frac{e^{-0.03y}}{-0.03} \right]_{2000}^{b} \\
&= 100k \left[(0 - 0) - \left(\frac{e^{-0.02(2000)}}{-0.02} - \frac{e^{-0.03(2000)}}{-0.03} \right) \right] \\
&= 100k \left(50e^{-40} - \frac{100}{3} e^{-60} \right)
\end{aligned}
$$

Substitute $k = 6 \times 10^{-6}$:

$$
\begin{aligned}
P(Y > 2000) &= 6 \times 10^{-4} \left(50e^{-40} - \frac{100}{3} e^{-60} \right) \\
&= 0.03e^{-40} - 0.02e^{-60}.
\end{aligned}
$$

5.2 Conditional Probability Distributions and Independence

5.2.1 Independence

Definition. Let X and Y be two random variables. Then they are **independent** if and only if

$$f_{XY}(x,\, y) = f_X(x)f_Y(y).$$

Remark. Recall math in high school, we know that

$$P(A \cap B) = P(A)P(B)$$

if both events A and B are independent.

Definition. Let X and Y be two random variables. Then the mean from a joint distribution $(X,\, Y)$ with a transformer function $H(X,\, Y)$ of X and Y is defined as

$$E[H(X,\, Y)] = \sum_{x \in X} \sum_{y \in Y} H(X,\, Y) f_{XY}(x,\, y) \qquad \text{(Discrete case)}$$

$$E[H(X,\, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} H(X,\, Y) f_{XY}(x,\, y)\, dx\, dy \qquad \text{(Continuous case)}$$

Example. Find $\displaystyle\sum_{x \in X} \sum_{y \in Y} x f_{XY}(x,\, y)$.

Sol.

$$\begin{aligned} \sum_{x \in X} \sum_{y \in Y} x f_{XY}(x,\, y) &= \sum_{x \in X} x \left(\sum_{y \in Y} f_{XY}(x,\, y) \right) \\ &= \sum_{x \in X} x f_X(x) \\ &= E(X). \end{aligned}$$

Example. Find $\displaystyle\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{XY}(x,\, y)\, dy\, dx$.

Sol.

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{XY}(x,\, y)\, dy\, dx &= \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f_{XY}(x,\, y)\, dy \right) dx \\ &= \int_{-\infty}^{\infty} x f_X(x)\, dx \\ &= E(X). \end{aligned}$$

5.2.2 Covariance and Correlation

Let's review the variance in one variable.

$$\begin{aligned} Var(X) &= E((X-E(X))^2) \\ &= E(X^2 - 2XE(X) + E^2(X)) \\ &= E(X^2) - 2E^2(X) + E^2(X) \\ &= E(X^2) - E^2(X) \end{aligned}$$

So we have

$$\begin{aligned} Var(X) &= E(X^2) - E^2(X) \\ &= E(X \cdot X) - E(X) \cdot E(X) \\ &\Rightarrow E(X \cdot Y) - E(X) \cdot E(Y). \end{aligned}$$

$E(X \cdot Y) - E(X) \cdot E(Y) = E[(X - E(X))(Y - E(Y))]$, it is covariance(共變異數).

Definition. The **covariance** between the random variables X and Y, denoted as $\text{cov}(X, Y)$ or σ_{XY} is

$$\text{cov}(X, Y) \equiv E[(X - E(X))(Y - E(Y))] = E(XY) - E(X)E(Y).$$

Theorem. Let (X, Y) be a ordered pair random variables with joint probability distribution function $f_{XY}(x, y)$. If X and Y are independent, then $\text{cov}(X, Y) = 0$.

Definition. (Correlation coefficient) The **Pearson correlation coefficient**(皮爾森相關係數) between random variables X and Y, denoted as ρ_{XY}, is

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{Var(X)Var(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

For any two random variables X and Y,

$$-1 \leq \rho_{XY} \leq 1.$$

Theorem. The conditions for **linearity** between X and Y if and only if

$$Y = \beta_0 + \beta_1 X.$$

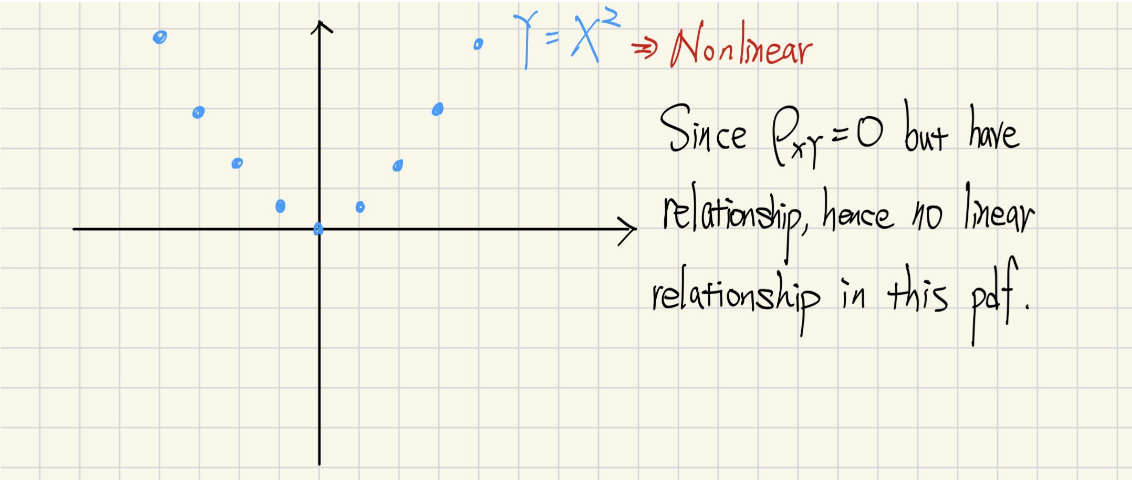
Remark. Let ρ_{XY} be the Pearson correlation coefficient between random variables X and Y.

1. If $\rho_{XY} = 1$, then we call it is perfect positive correlation and $\beta_1 > 0$ for $Y = \beta_0 + \beta_1 X$.

2. If $\rho_{XY} = -1$, then we call it is perfect negative correlation and $\beta_1 < 0$ for $Y = \beta_0 + \beta_1 X$.

3. If $\rho_{XY} = 0$, then we call it is no linear relationship.

Example. Consider $Y = X^2$.



Remark. If X and Y are independent, then $\rho_{XY} = 0$.

5.2.3 Conditional Probability Distributions

Definition. Let (X, Y) be a ordered pair random variables with joint probability distribution function $f_{XY}(x, y)$. Suppose $f_X(x)$ and $f_Y(y)$ are marginal probability distribution function for $f_{XY}(x, y)$.

(1) The conditional density for X given $Y = y$ is defined as

$$f_{X|y}(x) = \frac{f_{XY}(x, y)}{f_Y(y)}.$$

Recall math in high school, we know that

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

(2) The conditional density for Y given $X = x$ is defined as

$$f_{Y|x}(y) = \frac{f_{XY}(x, y)}{f_X(x)}.$$

Example. Suppose

$$f_{XY}(x,\,y) = \frac{c}{x}, \quad c = \frac{1}{6 - 27\ln{(33/27)}},$$

where $27 \le y \le x \le 33$. Find

(a) $f_X(x)$

(b) $f_Y(y)$

(c) $f_{X|y}(x)$

(d) The probability that $x > 32$ given $y = 30$.

(e) The expected value of X given $y = 30$.

Sol. We don't substitute c into the function since it is too difficult to evaluate.

(a)

$$\begin{aligned} f_X(x) &= \int_{27}^{x} f_{XY}(x,\,y)\,dy \\ &= \int_{27}^{x} \frac{c}{x}\,dy \\ &= \frac{cy}{x}\Big|_{27}^{x} \\ &= c\frac{x-27}{x}, \text{ where } 27 \le x \le 33. \end{aligned}$$

(b)

$$\begin{aligned} f_Y(y) &= \int_{y}^{33} f_{XY}(x,\,y)\,dx \\ &= \int_{y}^{33} \frac{c}{x}\,dx \\ &= c(\ln 33 - \ln y) \\ &= c\ln\frac{33}{y}, \text{ where } 27 \le y \le 33. \end{aligned}$$

(c)

$$\begin{aligned} f_{X|y}(x) &= \frac{f_{XY}(x,\,y)}{f_Y(y)} \\ &= \frac{c/x}{c(\ln 33 - \ln y)} \\ &= \frac{1}{x(\ln 33 - \ln y)} \\ &= \frac{1}{x\ln(33/y)}, \text{ where } y \le x \le 33. \end{aligned}$$

(d)

$$\begin{aligned}
P(X > 32|y = 30) &= \int_{32}^{33} \frac{1}{x(\ln 33/30)}\, dx \\
&= \left[\frac{\ln|x|}{\ln(33/30)}\right]_{32}^{33} \\
&= \frac{\ln 33 - \ln 32}{\ln(33/30)} \\
&= \frac{\ln(33/32)}{\ln(33/30)}.
\end{aligned}$$

(e)

$$\begin{aligned}
E(X|y = 30) &= \int_{30}^{33} x f_{X|y=30}(x)\, dx \\
&= \int_{30}^{33} x \frac{1}{x \ln(33/30)}\, dx \\
&= \int_{30}^{33} \frac{1}{\ln(33/30)}\, dx \\
&= \frac{3}{\ln(33/30)}.
\end{aligned}$$

5.3 Joint Probability Distributions for More Than Two Random Variables

Definition. A **joint probability density function** for the continuous random variables X_1, X_2, ..., X_p, denoted as $f_{X_1X_2\cdots X_p}(x_1,\, x_2,\, \ldots,\, x_p)$, satisfies the following properties:

(1) $f_{X_1X_2\cdots X_p}(x_1,\, x_2,\, \ldots,\, x_p) \geq 0$

(2) $\displaystyle\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} f_{X_1X_2\cdots X_p}(x_1,\, x_2,\, \ldots,\, x_p)\, dx_1\, dx_2 \ldots dx_p = 1$

(3) For any region B of p-dimensional space,

$$P((x_1,\, x_2,\, \ldots,\, x_p) \in B) = \int\cdots\int_B f_{X_1X_2\cdots X_p}(x_1,\, x_2,\, \ldots,\, x_p)\, dx_1 \ldots dx_p$$

Definition. The marginal density of X_1 for joint probability density function

$$f_{X_1X_2\cdots X_p}(x_1,\, x_2,\, \ldots,\, x_p)$$

is defined as

$$f_{X_1}(x_1) = \int\cdots\int_{\bigcup\limits_{k=2}^{p} X_k} f_{X_1X_2\cdots X_p}(x_1,\, x_2,\, \ldots,\, x_p)\, dx_2\, dx_3 \ldots dx_p.$$

Definition. From above definition, we can define mean and variance that is

$$\begin{aligned} E(X_1) &= \int\cdots\int_{\bigcup\limits_{k=2}^{p} X_k} x_1 f_{X_1X_2\cdots X_p}(x_1,\, x_2,\, \ldots,\, x_p)\, dx_2\, dx_3 \ldots dx_p \\ &= \int_{-\infty}^{\infty} x_1 f_{X_1}(x_1)\, dx_1 \end{aligned}$$

and

$$Var(X_1) = \int_{-\infty}^{\infty} [x_1 - E(X_1)]^2 f_{X_1}(x_1)\, dx_1.$$

Remark. Conditional probability distributions can be developed for multiple random variables by an extension of the ideas used for two random variables. for example, the joint conditional probability distribution of X_1, X_2 and X_3 given $X_4 = x_4$ and $X_5 = x_5$ is

$$f_{X_1X_2X_3|x_4x_5}(x_1,\, x_2,\, x_3) = \frac{f_{X_1X_2X_3X_4X_5}(x_1,\, x_2,\, x_3,\, x_4,\, x_5)}{f_{X_4X_5}(x_4,\, x_5)}$$

for $f_{X_4X_5}(x_4,\, x_5) > 0$.

Example. Suppose that the random variables X, Y and Z have joint probability density function

$$f_{XYZ}(x,\, y,\, z) = c$$

over the cylinder $x^2 + y^2 < 4$ and $0 < z < 4$.

(1) Determine the constant c so that $f_{XYZ}(x,\, y,\, z)$ is a probability distribution function.

(2) Find $P(X^2 + Y^2 < 2)$

(3) Find $P(Z < 2)$

(4) Find $E(X)$

(5) Find $P(X < 1|Y = 1)$

(6) Find $P(X^2 + Y^2 < 1|Y = 1)$

(7) Find the conditional probability distribution of Z given that $X = 1$ and $Y = 1$.

Sol.

(1) We want to find c such that $\iiint_T c\, dx\, dy\, dz = 1$.
Since we know that it is over the cylinder $x^2 + y^2 < 4$ and $0 < z < 4$.
Let $T = \{(x,\, y,\, z)\,|\, x^2 + y^2 < 4,\, 0 < z < 4\}$. This implies

$$\begin{aligned}\iiint_T c\, dx\, dy\, dz &= c\iiint_T 1\, dx\, dy\, dz \\ &= c\iiint_T 1\, dV \\ &= c \times 2^2 \times 4 \times \pi \\ &= 16\pi \times c = 1\end{aligned}$$

Hence $c = \dfrac{1}{16\pi}$.

(2) Let $T = \{(x,\, y,\, z)\,|\, 0 < z < 4,\, -\sqrt{2} < y < \sqrt{2},\, -\sqrt{2-y^2} < x < \sqrt{2-y^2}\}$.

$$\begin{aligned}P(X^2 + Y^2 < 2) &= \iiint_T c\, dx\, dy\, dz \\ &= c\int_0^4 \int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-y^2}}^{\sqrt{2-y^2}} 1\, dx\, dy\, dz \\ &= c\int_0^4 dz \int_{-\sqrt{2}}^{\sqrt{2}} 2\sqrt{2-y^2}\, dy \\ &= 8c\int_{-\sqrt{2}}^{\sqrt{2}} \sqrt{2-y^2}\, dy \\ &= 8 \times \frac{1}{16\pi} \times \pi = \frac{1}{2}.\end{aligned}$$

(3)

$$P(Z < 2) = c \iiint_T dV \times \frac{1}{2} = \frac{1}{2}.$$

(4) Let $T = \{(x,\, y,\, z) \,|\, 0 < z < 4,\, -\sqrt{4-x^2} < y < \sqrt{4-x^2},\, -2 < x < 2\}$.

$$\begin{aligned} E(X) &= \int_{-2}^{2} x f_X(x)\, dx \\ &= \int_{-2}^{2} x \left[\int_0^4 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} c\, dy\, dz \right] dx \\ &= 4c \int_{-2}^{2} x \cdot 2\sqrt{4-x^2}\, dx \\ &= 8c \int_{-2}^{2} x\sqrt{4-x^2}\, dx \\ &= 8c \left[\frac{-1}{3} u^{3/2} \right]_0^0 = 0. \end{aligned}$$

(5) Let $R = \{(x,\, z) \,|\, 0 < z < 4,\, -\sqrt{4-y^2} < x < \sqrt{4-y^2}\}$.
Compute

$$\begin{aligned} f_{XZ|y=1}(x,\, z) &= \frac{c}{f_Y(y=1)} \\ &= c \Big/ \iint_R c\, dx\, dz \Big|_{y=1} \\ &= \left[\int_0^4 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} 1\, dx\, dz \right]_{y=1}^{-1} \\ &= \left[\int_0^4 \int_{-\sqrt{3}}^{\sqrt{3}} 1\, dx\, dz \right]^{-1} \\ &= \frac{1}{8\sqrt{3}} \end{aligned}$$

Since it is given $Y = 1$, we have

$$x^2 + 1 < 4 \iff -\sqrt{3} < x < \sqrt{3}.$$

Suppose $\Omega = \{(x,\, z) \,|\, -\sqrt{3} < x < 1,\, 0 < z < 4\}$.
And thus

$$\begin{aligned} P(X < 1 | Y = 1) &= \iint_\Omega f_{XZ|y=1}(x,\, z)\, dx\, dz \\ &= \int_0^4 \int_{-\sqrt{3}}^{1} \frac{1}{8\sqrt{3}}\, dx\, dz \\ &= 4 \times \frac{1+\sqrt{3}}{8\sqrt{3}} = \frac{1+\sqrt{3}}{2\sqrt{3}}. \end{aligned}$$

(6) Given $x^2 + y^2 < 1$ and $Y = 1$, we obtain $x^2 + 1 < 1 \iff x^2 < 0$. Since there is no real number x satisfying this inequality, the probability is 0. Hence

$$P(X^2 + Y^2 < 1|Y = 1) = 0.$$

(7) Compute

$$P_{Z|x=1,\, y=1}(z) = \frac{c}{f_{XY}(x = 1,\ y = 1)} = \frac{1}{4}.$$

5.4 Common Joint Distributions

5.4.1 Multinomial Probability Distribution

Remark. Recall binomial distribution, let X be the number of trials that result in a success $x = 0, 1, \ldots, n$, $n = 1, 2, 3, \ldots$ with the parameter $p \in (0, 1)$. Then

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}.$$

Definition. Suppose that a random experiment consists of a series of n trials. Assume that

(1) The result of each trial is classified into one of k classes.

(2) The probability of a trial generating a result in class 1, class 2,$\cdots$, class k is $p_1, p_2, \ldots, p_k$, respectively, and these probabilities are constant over the trials.

(3) The trials are independent.

The random variables $X_1, X_2, \ldots, X_k$ hat denote the number of trials that result in class 1, class 2,$\cdots$, class k, respectively, have a **multinomial distribution**(多項分布) and the joint probability mass function is

$$P(X_1 = x_1, X_2 = x_2, \ldots, X_k = x_k) = \frac{n!}{x_1! x_2! \cdots x_k!} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k}$$

for $x_1 + x_2 + \cdots + x_k = n$ and $p_1 + p_2 + \cdots + p_k = 1$.

Theorem. If $X_1, X_2, \ldots, X_k$ have a multinomial distribution, the marginal probability distribution of X_i is binomial with

$$E(X_i) = np_i$$

and

$$V(X_i) = np_i(1 - p_i).$$

5.4.2 Bivariate Normal Probability Density Function

Remark. Recall normal distribution, we know

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where $-\infty < x < \infty$.

Definition. The probability density function of a **bivariate normal distribution**(**二變數常態分佈**) is

$$f_{XY}(x,\, y,\,) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}e^{u}$$

for

$$u = \frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right],$$

where

$$\begin{aligned} -\infty < x < \infty, &\qquad -\infty < y < \infty, \\ \sigma_X > 0, &\qquad -\infty < \mu_X < \infty, \\ \sigma_Y > 0, &\qquad -\infty < \mu_Y < \infty, \\ -1 < \rho < 1. & \end{aligned}$$

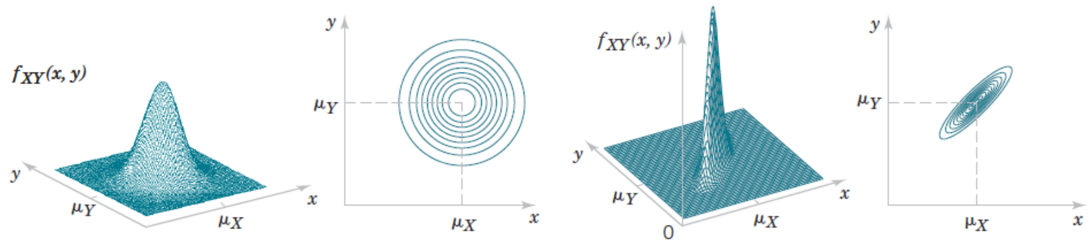


FIGURE 5.12

Examples of bivariate normal distributions.

Theorem. Suppose X and Y are bivariate normal distribution. If $\rho = 0$, then X and Y are independent.

In other view, we can use a matrix to define bivariate normal distribution and even more generalizing.

Definition. The probability density function of a multivariate normal distribution is

$$f_X(x_1,\, \ldots,\, x_k) = \frac{1}{\sqrt{(2\pi)^k|\Sigma|}}e^{\left(-\frac{1}{2}(\mathbf{x}-\mu)^T\Sigma^{-1}(\mathbf{x}-\mu)\right)}$$

for Σ is a covariance matrix of $X_1,\, \ldots,\, X_k$.

Remark. In bivariate case, it is defined

$$\mu = \begin{pmatrix}\mu_X \\ \mu_Y\end{pmatrix}, \quad \Sigma = \begin{pmatrix}\sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2\end{pmatrix}.$$

5.5 Linear Function of Random Variables

Definition. Suppose $X_1, X_2, \ldots, X_p$ are random variables and $c_1, c_2, \ldots, c_p \in \mathbb{R}$. Then the linear combination

$$Y = c_1X_1 + c_2X_2 + \cdots + c_pX_p$$

is a linear function of $X_1, X_2, \ldots, X_p$.

Remark. If $Y = c_1X_1 + c_2X_2 + \cdots + c_pX_p$, then

(1) $E(Y) = c_1E(X_1) + c_2E(X_2) + \cdots + c_pE(X_p)$

(2)

$$\begin{aligned} Var(Y) =& c_1^2Var(X_1) + c_2^2Var(X_2) + \cdots + c_p^2Var(X_p) \\ &+ 2\sum\sum_{i<j} c_ic_j\text{cov}(X_i, X_j) \end{aligned}$$

Theorem. If $X_1, X_2, \ldots, X_p$ are independent such that $\text{cov}(X_i, X_j) = 0$ for $i \neq j$, then

$$Var(Y) = c_1^2Var(X_1) + c_2^2Var(X_2) + \cdots + c_p^2Var(X_p).$$

Remark. Let $\bar{X} = \dfrac{X_1 + X_2 + \cdots + X_p}{p}$ be the average of random variable.

(1) $E(\bar{X}) = \dfrac{1}{p}E\left(X_1 + X_2 + \cdots + X_p\right)$, if X_i are independent and $E(X_i) = \mu$, we obtain $E(\bar{X}) = \mu$.

(2) Suppose X_i are independent and $Var(X_i) = \sigma^2$, then

$$Var(\bar{X}) = \frac{\sigma^2}{p}.$$

Proof. (1) Suppose X_i are independent and $E(X_i) = \mu$, thus

$$\begin{aligned} E(\bar{X}) &= \frac{1}{p}E\left(X_1 + X_2 + \cdots + X_p\right) \\ &= \frac{1}{p}\sum_{k=1}^{p} E(X_k) \\ &= \frac{1}{p}\sum_{k=1}^{p} \mu = \frac{1}{p} \times p\mu = \mu. \end{aligned}$$

(2)

$$Var(\bar{X}) = \frac{1}{p^2}\sum_{k=0}^{p} Var(X_k) = \frac{1}{p^2} \cdot p\sigma^2 = \frac{\sigma^2}{p}.$$

$\square$

5.6 General Functions of Random Variables

Let $y = f(x)$, we can define a transformation from X to Y.

Definition. Suppose that X is a **discrete** random variable with probability distribution $f_X(x)$. Let $Y = h(X)$ define a one-to-one transformation between the values of X and Y so that the equation $y = h(x)$ can be solved uniquely for x in terms of y. Then the probability mass function of the random variable Y is

$$f_Y(y) = f_X[u(y)].$$

Example. Let X be a geometric random variable with pmf

$$f_X(x) = p(1-p)^{x-1},\ x = 1,\ 2,\ \dots.$$

Find the pmf of $Y = X^2$.

Sol. Let $f_Y(y) = P(Y = y)$, since $Y = X^2$, it implies $X = \sqrt{Y}$. Thus

$$P(Y = y) = P(X = \sqrt{y}).$$

Because $P(X = x) = f_X(x) = p(1-p)^{x-1}$, $x = 1,\ 2,\ \dots$, we have

$$P(Y = y) = P(X = \sqrt{y}) = p(1-p)^{\sqrt{y}-1} = f_Y(y).$$

Hence

$$f_Y(y) = p(1-p)^{\sqrt{y}-1},\ y = 1^2,\ 2^2,\ 3^2,\ \dots$$

Suppose that X is a **continuous** random variable with probability distribution $f_X(x)$. Let $Y = g(X)$ be a random variable and $g(x)$ is strictly monotonic and differentiable. Since $F'_Y(y) = f_Y(y)$, it is clear that $F_Y(y) = P(Y \le y)$. Therefore, because $Y = g(X)$ and $g^{-1}(Y)$ exists, assume g^{-1} is decreasing, we obtain $P(Y \le y) = P(g(x) \le y)$. Then

$$\begin{aligned} F_y(y) &= P(g(x) \le y) \\ &= P(g^{-1}(g(x)) \le g^{-1}(y)) \\ &= P(X \ge g^{-1}(y)) \\ &= 1 - P(X \le g^{-1}(y)) \\ &= 1 - F_X(g^{-1}(y)) \end{aligned}$$

Differentiate $F_y(y)$, we have

$$\begin{aligned} F'_Y(y) &= 0 - f_X(g^{-1}(y))\frac{dg^{-1}(y)}{dy} \\ &= -f_X(g^{-1}(y)) \cdot (-1)\left|\frac{dg^{-1}(y)}{dy}\right| \\ &= f_X(g^{-1}(y))\left|\frac{dg^{-1}(y)}{dy}\right|. \end{aligned}$$

Theorem. Given X is a **continuous** random variable with probability distribution $f_X(x)$. Let $Y = g(X)$ be a random variable and $g(x)$ is strictly monotonic and differentiable. Then

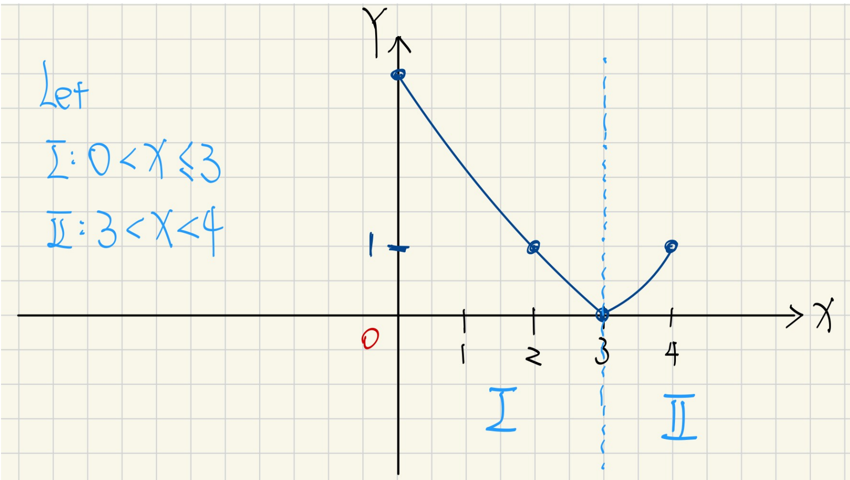
$$f_Y(y) = f_X(g^{-1}(y))\left|\frac{dg^{-1}(y)}{dy}\right|.$$

Remark. In continuous, $f_X(x) \neq P(X = x)$. (不一定, 根據連續的 cdf 和 pdf 的特性)

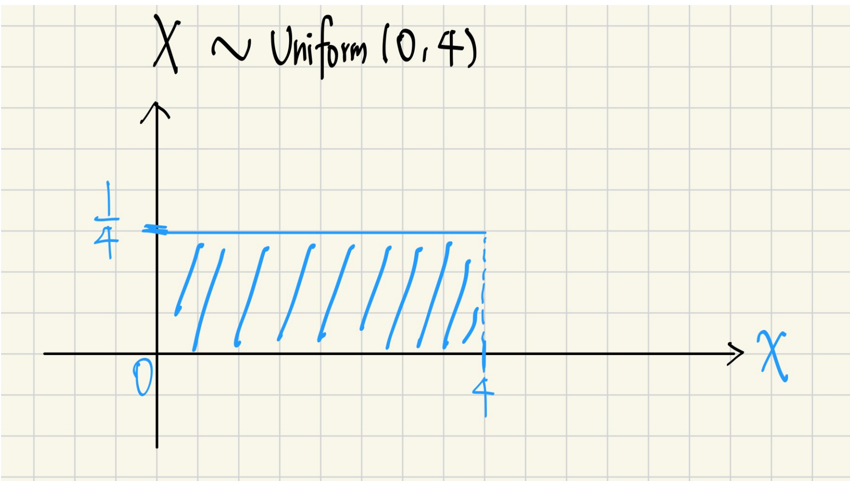
Example. Let X be uniformly distributed RV over $(0,\ 4)$. Let $Y = g(X) = (X-3)^2$. Find the pdf of Y.

Sol. Consider:

(1) $Y = (X-3)^2 = g(X)$



(2) Since it is uniform, we have



Hence

$$f_X(x) = \begin{cases} \frac{1}{4}, & 0 < x < 4 \\ 0, & \text{otherwise} \end{cases}$$

(I) For $0 < X \leq 3$, $Y = (X-3)^2 \implies Y = g(X)$

$$\begin{aligned} \pm\sqrt{Y} = X - 3 &\iff X = 3 \pm \sqrt{Y} \\ &\iff X = 3 - \sqrt{Y} = g^{-1}(Y). \end{aligned}$$

(II) For $3 < X < 4$, it is

$$X = 3 + \sqrt{Y} = g^{-1}(Y).$$

Using

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right|,$$

we have

(I) For $0 < X \leq 3$,

$$\begin{aligned} f_{Y_I}(y) &= f_X(3 - \sqrt{Y}) \left| \frac{d}{dy}(3 - \sqrt{Y}) \right| \\ &= \frac{1}{4} \cdot \left| -\frac{1}{2\sqrt{Y}} \right|, \ 0 < y < 9. \end{aligned}$$

(II) For $3 < X < 4$,

$$\begin{aligned} f_{Y_{II}}(y) &= \frac{1}{4} \cdot \left| \frac{d}{dy}(3 + \sqrt{Y}) \right| \\ &= \frac{1}{4} \cdot \left| \frac{1}{2\sqrt{Y}} \right|, \ 0 < y < 1. \end{aligned}$$

Hence

$$\begin{aligned} f_Y(y) &= \begin{cases} f_{Y_I}(y) + f_{Y_{II}}(y), & 0 < y < 1 \\ f_{Y_I}(y), & 1 < y < 9 \end{cases} \\ &= \begin{cases} \dfrac{1}{4\sqrt{y}}, & 0 < y < 1 \\ \dfrac{1}{8\sqrt{y}}, & 1 < y < 9. \end{cases} \end{aligned}$$

5.7 General Functions of Bivariate Random Variables

Let X and Y be random variables, and defined transformation

$$\begin{cases} U = g(X, Y) \\ V = h(X, Y). \end{cases}$$

Then the inverse transformation is

$$\begin{cases} X = h_1(U, V) \\ Y = h_2(U, V). \end{cases}$$

Given $f_{XY}(x, y)$, then we define

$$f_{UV}(u, v) = f_{XY}(h_1, h_2)|J|$$

where $J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \neq 0$ is a Jacobian.

Example. Consider the following transformation

$$T : \begin{cases} u = 3x + 2y \\ v = x - y \end{cases}$$

(a) Is the transformation invertible? If yes, find T^{-1}.

(b) Find the Jacobian for T^{-1}.

Sol. (a) Compute T^{-1}, we have

$$T^{-1} : \begin{cases} x = \dfrac{u + 2v}{5} \\ y = \dfrac{u - 3v}{5} \end{cases}$$

(b) Compute

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{1}{5} & \frac{-3}{5} \end{vmatrix}$$
$$= \frac{1}{25}(-3 - 2) = \frac{-1}{5}$$

Example. Let X and Y be independent uniformly distributed random variables over the intervals $(0,\, 2)$ and $(0,\, 3)$, respectively. Let $U = XY$ and $V = Y$, i.e.

$$T : \begin{cases} u = xy \\ v = y \end{cases}$$

(a) Find $f_X(x)$, $f_Y(y)$ and $f_{XY}(x,\, y)$.

(b) Find Jacobian.

(c) Find $f_{UV}(u,\, v)$

Sol. (a) Since $X \sim \text{uniform}(0,\, 2)$ and $Y \sim \text{uniform}(0,\, 3)$, therefore

$$\begin{cases} f_X(x) = \frac{1}{2}, & 0 < x < 2 \\ f_Y(y) = \frac{1}{3}, & 0 < y < 3. \end{cases}$$

Since X and Y are independent, then

$$f_{XY}(x,\, y) = \frac{1}{6},\ 0 < x < 2,\ 0 < y < 3.$$

(b)

$$T : \begin{cases} u = xy \\ v = y \end{cases} \quad \Longrightarrow T^{-1} : \begin{cases} x = \frac{u}{v} \\ y = v \end{cases}$$

Compute

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{v} & \frac{-u}{v^2} \\ 0 & 1 \end{vmatrix} = \frac{1}{v},\ 0 < v < 3.$$

(c) By the formula $f_{UV}(u,\, v) = f_{XY}(h_1,\, h_2)|J|$, hence

$$\begin{aligned} f_{UV}(u,\, v) &= \frac{1}{6} \cdot \left|\frac{1}{v}\right| \\ &= \frac{1}{6v},\ 0 < u < 2v,\ 0 < v < 3 \end{aligned}$$

By

$$\begin{aligned} 0 < x < 2 &\iff 0 < \frac{u}{v} < 2 \\ &\iff 0 < u < 2v \\ &\iff \left(0 < \frac{u}{2} < v < 3\right). \end{aligned}$$